Progress on Attention

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We report some developing work on the Anthropic interpretability team, which might be of interest to research working actively in this space. We'd ask you to treat these results like those of a colleague sharing some thoug preliminary experiments for a few minutes at a lab meeting, rather than a mature paper.

Last year, we announced that studying attention superposition was one of our team's priorities, motivated i a toy model demonstrating attention features in superposition. Since then, we've tried several ideas to stud problem in real language models, and we'd like to give an update on what we've tried and where we think t problem is now.

At a high level, we found significant evidence of attention superposition, and additionally of cross-layer atte representations. We looked for and did not find evidence of preferred OV dimensions for attention features further found evidence that the QK condition ("where is information moving to/from?") and the OV-conditi information is being moved?") are coupled, and have found a practical method for studying these in the for *Multi-Token Transcoders*.

We believe that the core remaining problem is understanding *why models attend where they do*, which is to understanding attention pattern formation. We believe we have a promising path forward on this problem, form of *QK diagonalization*, which we report on below.

Factoring the Problem

We want to understand the nature of computation in attention. To get a handle on this problem, we've foun useful to decompose it into a number of smaller pieces, each of which would either inform further investiga else provide a practical tool for analysis if solved. Those pieces are:

1. To what extent is there superposition across attention heads within a layer?
2. To what extent can we combine the QK-condition ("where is information moving to/from?") and the condition ("what information is being moved?") into individual attention features? Are these naturall coupled?
3. To what extent are attention features multidimensional in the OV circuit (e.g. $\mathrm{Rank}(W_{\mathrm{OV}}) > 1$)?

4. To what extent are attention features multidimensional in the QK circuit (e.g. $\mathrm{Rank}(W_{\mathrm{QK}}) > 1$)?

5. To what extent are attention features spread across layers?

6. How are attention patterns formed / how can we best understand the nonlinear action of the QK cir

We began our study with an ambitious approach that, if successful, would have answered most of these in o We then conducted several more targeted studies to solve individual pieces, and we now have preliminary r several of these.

In the rest of this update we document our approaches to date, explain how each has updated us, and prov roadmap of next steps.

Approaches

ATTENTION REPLACEMENT LAYER

Our first approach was an "Attention Replacement Layer". The basic idea was to form a wide (many-head) at layer and train it to mimic the behavior of an attention layer in the base model of interest (i.e. given the sam train on MSE versus the outputs of the base layer). The wide model would be regularized to be simple, in th the QK and OV circuits being low-rank. In some variants we also regularized to make the attention pattern s With this approach, an attention feature is a single head in the replacement layer.

We made some progress with this approach, including some signs of life (interpretable heads in the replace layer). At the same time, we hit significant ML difficulties that we were unable to resolve. Our current view is likely possible to get this approach to work, but not easy.

We also made some conceptual progress pursuing this approach. We studied a wide array of different regul schemes, and found that a regularization penalty of the form $[L_{\mathrm{nuc}}(W_{\mathrm{QK}})L_{\mathrm{nuc}}(W_{\mathrm{OV}})]^{\beta}, 1/2 < \beta < 1$ ha the theoretical properties we want in a simplicity regularizer[2] , including:

1. If the base model implements a set of independent skip-trigrams, this penalty incentivizes splitting t separate heads, each with rank-1 OV and QK circuits.
2. If the base model implements an induction feature, where a rank-N QK circuit performs prefix-match a rank-M OV circuit moves information, and if these circuits act in mostly-disjoint subspaces, the per incentivizes representing this as a single head with rank-N in the QK circuit and rank-M in the OV cir

We found some preliminary signs of life with this approach. In particular, we were able to identify heads in t replacement layer implementing specific skip-trigrams and we found heads that appeared to be performing induction. The results were always somewhat messy though, and we spent a fair bit of time trying to unders

In the end our most likely hypothesis is that the ML of our setup was not well-optimized. Unfortunately beca replacement layer has a lot of moving parts and was more-different from models we had worked with previ found it difficult to improve the ML. This was both for engineering reasons (scaling this approach to large nu heads efficiently proved challenging) and because there was significant design freedom (multiple hyperpara for the loss, multiple choices of loss functions, etc.) and no clear quantitative metric for comparing these.

As a result of these challenges, our approach now is focused on getting signal on different aspect of the prc isolation through a variety of simpler experiments.

MULTITOKEN TRANSCODERS

To get at questions (1,2,3), we trained Multitoken Transcoders (MTC's). An MTC takes as input the residual st over a full context at the input to a layer in the base transformer and predicts either the residual stream afte layer over the full context *or* the output of the attention part of the layer over the full context. It does this w components:

1. A token-by-token encoder on the source position, much like the encoder in an SAE. This produces s side feature activations.
2. An attention operation. For each feature we have a vector specifying a linear combination of attentic patterns from heads in the base layer, padded with an identity pattern to imitate the residual connec facilitate modeling of MLP computation, if desired. We call this the **head loading vector**, and we us form a feature-specific attention pattern, which we then apply to the source-side activations to mov activations to their destinations[3] .
3. A decoder on the destination position.

An advantage of this architecture is that it can implement a powerful fused Attention+MLP operation (with t identity like "attention head" allowing MLP-like computation to be learned), where information is moved an simultaneously run through an MLP, which allows MTC's to potentially produce simpler representations of c merging Attention-like operations with MLP-like operations that come either before or after the informatior moved[4]

Additionally, MTC's form attention patterns in the same way as our toy model of attention superposition, wh makes their performance something of a proxy for the hypothesis of attention superposition.

We train our MTC's on MSE relative to the objective, plus an L1 penalty on the encoder activations and a diff penalty on the negative pre-activations, which we've found to be an easy and effective way to avoid dead fe

With MTC's we are explicitly not studying QK dimensionality (question 4), cross-layer behavior (question 5), attention patterns form (question 6), since we've restricted ourselves to a single layer and in some sense inh attention patterns from that layer.

We trained MTC's with 100,000 features on an 18-layer transformer. We found that:

1. Many features show clean & interpretable attention patterns. As an example, we found a feature tha to outlying/surprising inputs (e.g. in a list of all 1's with a few 2's sprinkled in, it attends to the 2's; in with repeating patterns it attends to the first place where the pattern breaks, etc.) We also found a " feature which tracks valence when negative sentiment is about to get negated -- the feature writes c positive value up until the "flip," and then negative value once the flip has occurred.
2. We see features that clearly correspond to induction behavior, and these have head loading vectors primarily rely on known induction heads. Moreover, induction features cluster strongly by their head vectors, and it is common to see several features with near-identical head loadings.
3. We further found evidence that clustering features by their head-loading vectors produces semantic meaningful clusters beyond the induction example. So for instance we found a cluster of features inv

identifying where the present token is in a sentence/clause, another likely involved in predicting deli and another involved in induction in code contexts.

4. We see features that attend to the present token and **negatively** to the previous token (which is allc since we use linear combinations of attention patterns). We're not sure what behavior this enables, b common motif. One possibility is that this could be an analog of "edge detectors" for features acros
5. When we train MTC's to target the residual stream after a layer (rather than just the output of the at layer), we see many features that purely attend to the present token, indicating that these are fully M and do not participate in information movement. When we train MTC's to target the attention outpu we still see some such features (indicating some purely local computation being done by attention h but many fewer.

ATTENTION SUPERPOSITION

To understand the extent to which attention features exist in superposition (spread across heads), we compa MTC's against a baseline where each feature is bound to and moved by a single attention head[6] . The single MTC's performed significantly worse on (L0, MSE), usually a factor of 2-4x worse on L0 at matching MSE. We inspected the features in these single-head MTC's and they looked qualitatively worse than those in regular particular they had a much higher incidence of attending between seemingly-unrelated tokens / tokens that conform to the pattern shown by most token-pairs where the feature is active.

The fact that restricting MTC features to a single head degrades both performance and interpretability was a moderate update for us towards attention superposition as a hypothesis, especially since MTC's form attent patterns in precisely the same way as our attention superposition toy model.

QK/OV COUPLING

We want to understand the extent to which the QK-condition ("where is information moving to/from?") and condition ("what information is being moved?") are naturally coupled.

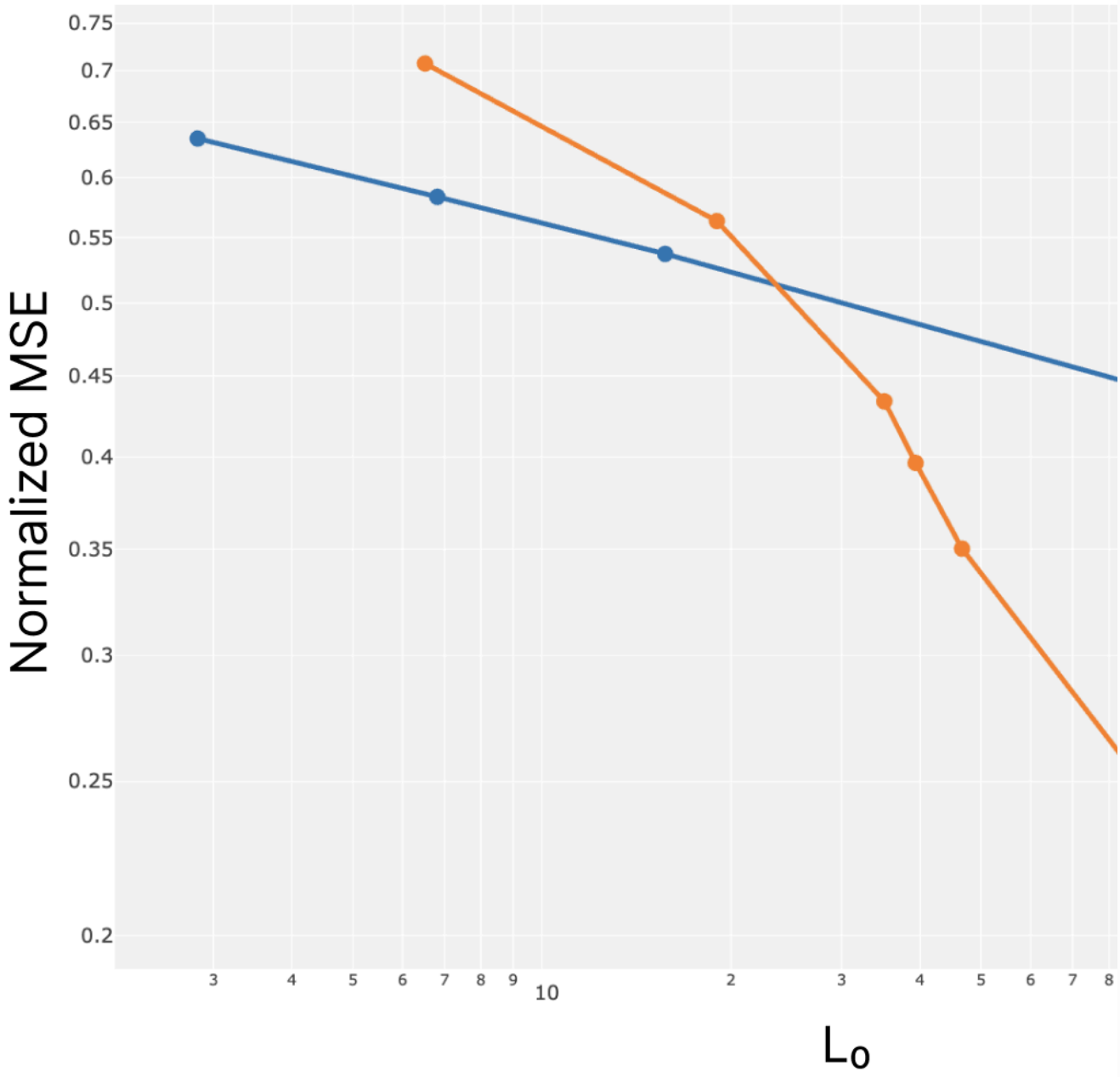
If these are coupled, we should expect MTC's to perform better than a baseline shaped like "move SAE featu through attention heads", since the difference between the two is that the MTC's can jointly learn what infor moving and how to move it, whereas the SAE doesn't get to see information about the information moveme

We weren't sure of exactly the right way to make this comparison, so we compared MTC's against a few diff baselines. First, we compared MTC features qualitatively against SAE features run through the single attentic that most-moves that feature in a given context. We made this comparison with the same contexts for both SAE features. Restricting the SAE features to use a single head favors the MTC in this comparison, but choos head dynamically according to the context favors the SAE, since the MTC cannot make that choice dynamica this setup, we found that the MTC features looked somewhat cleaner and more interpretable, but not hugel which we interpret as a weak update towards the QK and OV conditions being coupled.

As a second baseline, and to get some quantitative signal, we trained a model jointly on an SAE target and a target. That is, we trained an encoder and a decoder as an SAE (with no attention operation), and we trainec different decoder as a Top-1 MTC. The SAE & MTC steps see the same dataset examples and are interleaved MTC training in this case detaches gradients on the encoder SAE, so the SAE is the sole determinant of featu activations. The idea in this setup is to see if we can pick better features for modelling attention by accounti attention in the training process, or if SAE features perform comparably-well.

We qualitatively inspected the features that come from this training process and found that they were subst harder to interpret than the features we see in either regular MTC's or regular Top-1 MTC's. The SAE feature through a single-head MTC decoder were often quite fuzzy (i.e. broadly-attending) or else polysemantic.

We also quantitatively compared this baseline to regular Top-1 MTC's, and found that the jointly-trained mc performed somewhat better in the low-L_0 high-MSE limit but comparably or substantially worse everywher



Overall, these qualitative and quantitative results suggest that the QK and OV conditions *are* likely coupled i and we should choose approaches to attention that allow us to study them jointly.

MULTIDIMENSIONAL OV

Above we found semantically-meaningful clusters of features in head-loading space, including clusters of in features, clusters of features involving delimiters in text, and clusters of features involving code contexts. Sin clustering in head-loading space means a shared attention pattern, the differences among these features ha in what information they move rather than in their patterns.

This raises a question: are these clusters evidence of attention features with many OV dimensions? For insta induction feature might have a large OV dimension so it can move a wide range of different pieces of inforn Such features would naturally appear as clusters in head-loading space since our MTC only allows one OV d per feature.

Since we know that different attention behaviors require different OV dimensions (e.g. one dimension for sk trigrams, versus many for induction/copying behavior), we might expect to see many small clusters correspc skip-trigrams and related behaviors, and a smaller number of large clusters corresponding to induction/cop related behaviors.

To study this, we clustered the head-loading vectors of our features using k-means clustering, sweeping ove Instead of seeing two distinct populations of clusters, we see a smooth, power-law distribution of cluster siz this remains true across many orders of magnitude of k.

This suggests that attention behaviors might form more of a continuum in OV-dimension. For instance, it m that there are broad induction features that move large chunks of the residual stream as well as narrow one move smaller chunks. And similarly there might be generalizations of skip-trigram behavior that involve mo than one dimension at a time.

One potential limitation of this analysis is that we trained each MTC on just one layer. It is conceivable that, a substantial amount of cross-layer attention superposition, our feature clusters might be "blurred" by being to a single layer. We address this limitation using multi-layer methods in the next section.

ATTENTION VALUE SAE'S

After studying MTC's, we turned to Attention Value SAE's, which are SAE's trained on the concatenated value of multiple attention heads.

CROSS-LAYER REPRESENTATIONS

We first used Attention Value SAE's to study cross-layer attention representations. Concretely, we trained th Attention Value SAE's on an 18-layer model: one on the concatenated value vectors of the heads in layer 4, the concatenated value vectors of the heads in layer 10, and one on the concatenation of the value vectors heads in both of these layers. We chose layers 4 and 10 because they contained the clearest induction head seemed interesting to study.

We compared the two-layer SAE to the two one-layer SAE's by adding the L_0's of the two single-layer runs and combining their normalized MSE's (summing and dividing by $\sqrt{2}$). Across a wide range of sparsity pena two-layer SAE performed better than the combination of the one-layer SAE's, achieving roughly half the L_0 normalized MSE or roughly 0.3 better normalized MSE at fixed L_0. This is significant evidence in our view o layer attention representations, and an update towards approaches that can capture properties of multiple l once.

MULTIDIMENSIONAL OV

In our study of MTC's we used feature clustering to study the distribution of OV dimensions of attentional fe One concern we had is that our results may have been distorted or "blurred" by the MTC being limited to a layer.

To address this, we repeated our analysis using an Attention Value SAE trained on two layers. We constructe loading vector for each SAE feature using the norm of the decoder vector (per head) as a measure of head l We then clustered these vectors as before, using k-means clustering and sweeping over k.

The resulting clusters look very similar to those we found with a single-layer MTC: we see a power-law distri cluster sizes and no evidence of discrete populations of OV dimensions.

ATTENTION OUTPUT SAES

Another strategy we have explored is simply training SAEs on the output of attention layers, as explored in t work. As in prior work, we observe that these features are broadly interpretable and include induction featur succession features. In addition, we have explored combining them with cross-layer transcoders (CLT's, whic MLP outputs) to compute attribution graphs that include attention information. This improves on our CLT-o attribution graphs by localizing attention computation to particular layers and context position[8] , and allow see a feature visualization for each attention-mediated step. We have found this moderately useful in makir attribution graphs more interpretable, but not as much as we expected. Attention output features often see to those found in a CLT-only attribution graph, or easily "guessable" from them.

Notably, attention output SAE features do not help us address the question of how attention patterns are fo addition, compared to MTCs, attention output features provide less information – attention output SAE feat no fixed assignment to attention heads (they receive variable-by-prompt head contributions), and we can o visualize their "query-side" activations (unlike MTC features where we can visualize both key-side and query activations). For these reasons, we generally find MTC features to be preferable. However, attention output S somewhat more convenient to train (in particular, we can train them using only one token position's activati time, unlike MTCs where we need to train on full contexts).

We have also explored cross-layer variants of attention output SAEs that integrate with cross-layer transcod architecture we have found useful for attribution graphs is as follows:

- Two kinds of features in each layer. Type 1 reads in from the residual stream at that layer. Type 2 rea from the attention output of that layer.
- Type 1 features are "strictly causal" and attempt to reconstruct all downstream MLP outputs and atte layer outputs

- Type 2 features are "weakly causal" and attempt to reconstruct their input (the attention output at th as well as all downstream attention outputs and MLP outputs

The reason Type 2 features need to be weakly causal is that attention layer outputs are not fully predictable preceding residual stream at a single token position. We find this architecture improves MSE/L0 and simplifi attribution graphs somewhat, compared to combining CLTs and plain attention output SAEs. We take this as evidence for cross-layer representations in attention outputs.

ATTENTION PATTERN FORMATION

Aside from the Attention Replacement Layer, none of the methods above can tell us, even in principle, abou attention patterns are formed. In many cases this is fine: we can interpret MTC features just by looking at ex they fire on and studying the resulting attention patterns. But in many cases the important question is *why* t attended where it did.

For instance, in answering a multiple choice question the model has to choose one of the answers to attend the interesting logic is involved in this choice! So we really do want a means to understand how attention pa form.

To this end, we are pursuing an idea we call "QK diagonalization". The basic premise is to route the query an vectors from a given attention head in the base model through separate encoders to form query-side and k features, respectively. The idea behind "diagonalization" is to then only consider the interaction between the query feature and the ith key feature. Each interaction corresponds to a "QK feature", which yields a n_{ctx} by matrix of attention scores. We are experimenting with both rank-1 QK features (where the query-side and ke encoders are vectors, and the corresponding attention score matrix for each feature is rank 1) and high-rank features (where the query-side and key-side encoders are matrices, their interaction is computed using a do instead of a scalar product, and the corresponding attention score matrix may be high-rank). We then sum t feature-wise attention scores across all features and softmax the result to obtain the reconstructed attentior for the given head. During training, we penalize the MSE between this pattern and the base model pattern; apply various sparsity penalties (separate L1 penalties on the query- and key-side activations; an L1 penalty product of the activations; and a negative pre-activation penalty on the query- and key-side features, as exp above). In followup experiments, we jointly trained on all attention heads within a given layer, allowing featu read from the keys/queries of multiple heads and write to the attention patterns of multiple heads: in these how much a feature writes to a given head is determined by a learned head weights matrix. We also conside variations of the setup in which we use the features to predict the attention patterns directly (i.e., a linear ve which we don't apply the softmax).

Our results so far have shown some signs of life, but using rank-1 QK features it has proved challenging to f setup that faithfully matches the base attention patterns (i.e., normalized MSE $\lessapprox$ 0.25) with reasonable L0 (< One interesting (yet puzzling) result we found is that in almost all variations we tried, we find a relatively sm $\mathcal{O}(1000)$) number of induction features that are doing nearly all of the work. The remaining features – and true regardless of the total number of features – contribute negligibly to the MSE (as measured by ablation experiments).In the single-head experiments, the non-induction features are usually completely dead, or so active that they don't meaningfully contribute to the MSE. In the multi-head experiments, if we don't place constraints on the sign of the head weights, we find that these features tend to *cancel out*; half contribute n attention scores and half contribute positive attention scores, resulting in net zero effect on the MSE.

This behavior – in the multi-head case in particular – is puzzling because the model is choosing to increase L without any improvement in MSE, again, regardless of the total number of features. Unlike when training autoencoders, increasing the total number of features here does not improve MSE (at fixed L0). This points a an architectural issue – perhaps it is just not possible to faithfully model attention pattern formation using a reasonable number of rank-1 QK features – or an ML issue in our training setup. We suspect it may be the fo and as a result we are now pursuing the high-rank QK feature approach.

That said, the induction features we *do* find using the rank-1 approach are typically quite interpretable and c significantly more complex than token-based induction. For instance, we have found features that mediate i on surnames, on syllables within words that start with a C or a K, or on words related to cardinal or spatial d

We have only been working on this approach in earnest for a short time, so we expect that there is more to learned by pushing more in this direction.

Conclusion and Outlook

For the first time it feels like we have real traction on understanding attention. In particular, we are now mor confident that there is attention superposition in models across both heads and layers. We have updated av thinking that attentional features form discrete families grouped by dimensionality. And we have developed practical method in the form of MTC's to answer the questions of "where is the information moving to/from "what information is being moved?".

Our biggest remaining question is how to understand *why models attend where they do*. If we had a solutior we could combine it with our solutions above to form a complete picture of information movement in mode

Given this, our next steps split into two classes: understanding pattern formation, and further developing ou above to improve our circuit analyses:

1. Attention Pattern Formation
    1. We intend to continue studying attention pattern formation through the lens of QK diagona and related methods.
1. Circuit Analysis
    1. We may decide to extend MTC's to be Multitoken Cross-layer Transcoders (MCTC's). If we di could potentially replace CLT's and Attention Output SAE's in our circuit analysis framework MCC's, which would simplify both training and analysis.
    2. Even if we do not extend MTC's to MCC's, we may decide to use MTC's in place of Attention SAE's in studying circuits.
    3. Even though we did not see signs of a *single* preferred dimensionality in MTC features, we th is likely value in studying MTC's with multi-dimensional information movement using a distr dimensionalities, analogous to attention heads with varying head dimensions $d_{\text{head}} > 1$.

We may also return to study attention replacement layers, likely through a lens that looks like a combinatior diagonalization and MTC's.

Footnotes

1. One might have expected to find that features either prefer to have one dimension, to encode a skip-trigram, or else a high dimension, to large chunks of information. Instead we see a continuous spectrum. [↩]

2. Computing the nuclear norm directly is expensive, but it turns out that there is a cheap proxy to this when regularizing factored matrices. particular, if $W_{\mathrm{QK}} = W_{\mathrm{Q}} W_{\mathrm{K}}$ then the minima of $L_2(W_{\mathrm{Q}})^2 + L_2(W_{\mathrm{K}})^2$ are the same as those of $L_{\mathrm{nuc}}(W_{\mathrm{QK}})$, and the same is true of functions of the nuclear norm. See Mazumder+2010 and Kunin+2019 for more details. [↩]

3. For performance reasons we only move the top-k features produced by the encoder on any given context, sorting by the summed activat across the context. This is technically an acausal operation if k is less than the number of active features, but we find that by choosing a la enough k we can ensure that the top-k selection only matters early in training, as fewer than k features are active on any given context la training. [↩]

4. Ameisen+2025 has a discussion of how transcoders can be used to study circuits in models. In this context, MTC's can be seen as extendin transcoders to incorporate information about attention. [↩]

5. The penalty on the negative pre-activations in some sense sets the desired mean feature density, since it is active on most examples and against the regular L1 penalty times the feature density, so the desired pre-activation penalty is often many orders of magnitude smaller t regular L1 penalty. [↩]

6. We implemented this by training an MTC with a Top-1 selection on top of the head loading vector. [↩]

7. It would also be interesting to compare the MTC features against SAE features moved through **all** attention heads, though we have not ye this. [↩]

8. CLT-only attribution graphs integrate across all possible paths through attention heads in all layers to compute edge strengths. [↩]